

7(a)	<i>any four from:</i> 1 haemoglobin less soluble ; 2 (if oxygen concentration decreases) haemoglobin molecules, stick together / form long fibres ; 3 red blood cells, pulled out of shape / become sickle shaped ; 4 blood poor at transporting oxygen ; 5 (so) less oxygen getting to, cells / tissues / organs ; 6 red blood cells may, get stuck in / block, capillaries / vessels ; 7 pain / sickle cell crisis / fatigue ;	4
7(b)(i)	both alleles, contribute to / expressed in, the phenotype (in the heterozygote) ;	1
7(b)(ii)	<i>any one from:</i> smaller quantity of abnormal haemoglobin ; fewer red blood cells become sickled / sickling less severe ; ORA not affected by low(er) oxygen concentration ;	1

Question	Answer					Marks
7(b)(iii)	parental phenotype	SCT		SCT		3
	parental genotype	$Hb^A Hb^S$		$Hb^A Hb^S$;		
	gametes	Hb^A	Hb^S	Hb^A	Hb^S	
	offspring genotypes	$Hb^A Hb^A$	$Hb^A Hb^S$	$(Hb^A Hb^S)$	$Hb^S Hb^S$;	
	offspring phenotypes	normal	SCT	(SCT)	SCA ;	
	<i>I carrier</i>					